AN OVERVIEW OF MICROBIALLY INDUCED CALCITE PRECIPITATION (MICP) TECHNOLOGY

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Introduction: What is MICP?

- A biogeochemical process - specific **microorganisms** facilitate the **precipitation of calcium carbonate (calcite)** from a solution.

- Some primary MICP mechanisms are
  - Urease bacteria
  - Anaerobic denitrifying bacteria
  - Sulphate-reducing bacteria
  - Iron-reducing bacteria

- **Urease-producing bacteria** - the most widespread and useful type.

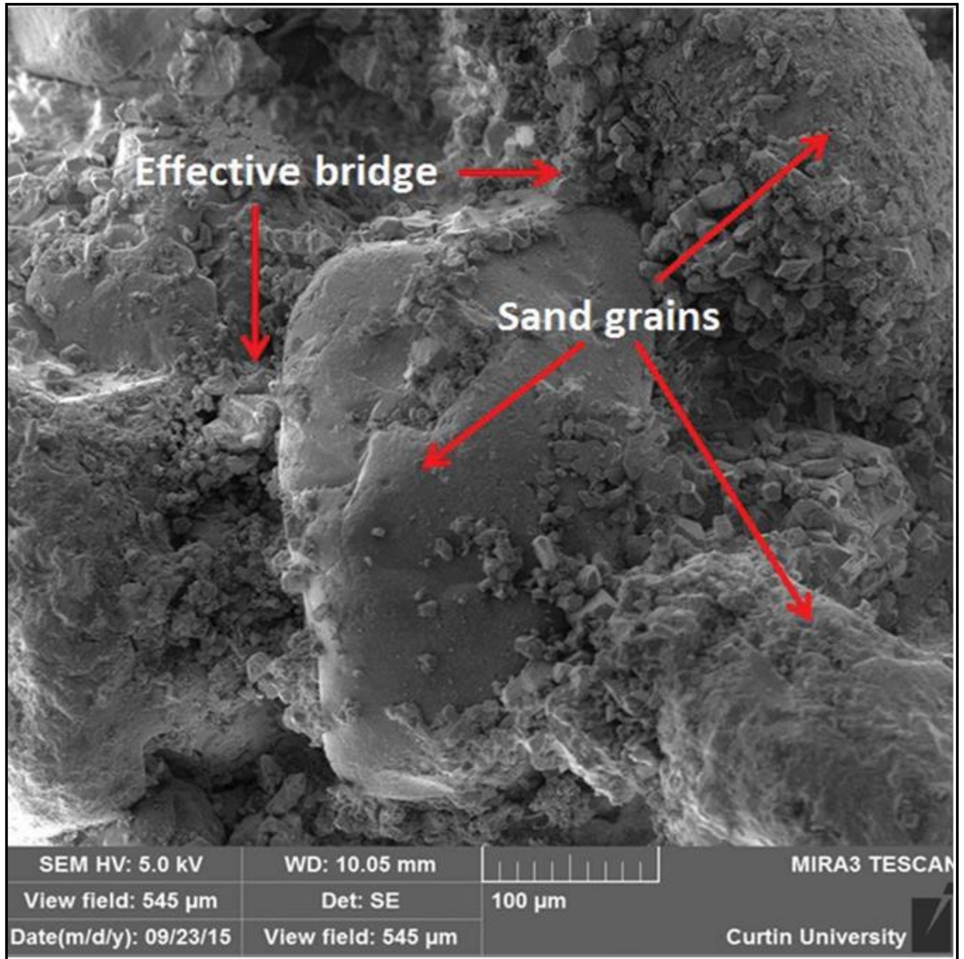

Fig.1 SEM image showing effective bridge formation (Mujah et al. 2016)

Mechanism of MICP

Table 1. Mechanism of MICP (Jiang et al. 2023)

Metabolic pathway	Biochemical reaction	Adaptation to environment	Representative bacterial type	Application area
Urease bacteria	$CO(NH_2)_2 + 2H_2O \rightarrow H_2CO_3 + 2NH_3$ $H_2CO_3 + 2NH_3 \rightarrow 2NH_4^+ + 2OH-$ $H_2CO_3 \rightarrow H^+ + HCO_3^-$ $HCO_3^- + H^+ + 2OH- \rightarrow CO_3^{2-} + 2H_2O$ $Ca^{2+} + CO_3^{2-} \rightarrow CaCO_3\downarrow$	Aerobic and alkaline environment	*Bacillus pasteurii, Sporosarcina pasteurii, Bacillus sphaericus, Bacillus subtilis, Bacillus megaterium, Bacillus diminuta* et al.	Soil reinforcement, treatment of liquefied soil, stable slope, repair of concrete cracks, biological bricks, dust prevention and sand fixation, heavy metal pollution control
Anaerobic denitrifying bacteria	$2.6Ca^{2+} + 2CH_3COO- + 3.2 NO_3^- \rightarrow 2.6CaCO_3\downarrow + 1.6 N_2 + 1.4CO_2$	Under anoxic and alkaline conditions	*Alcaligenes, Spirillum, Pseudomonas, Achromobacter, Serratia marcescens, Diaphorobacter nitroreducens.et* al.	Soil reinforcement, repair of concrete cracks, heavy metal pollution control
Sulphate-reducing bacteria	$2CH_2O + OH- + SO_4^{2-} \rightarrow HS- + H_2O + 2 HCO_3^-$ $HCO_3^- \rightarrow OH- + CO_2$ $Ca^{2+} + HCO_3^- + OH- \rightarrow CaCO_3\downarrow + H_2O$	Anaerobic, alkaline environment	*Desulfovibrio, Desulfobacilli, Desulfurmonas, Desulfobulbus, Desulfotomaculum* et al.	Soil reinforcement, anti-seepage plugging
Iron-reducing bacteria	$(HCOO)_3Fe + 3OH- + 3NH4^+ \rightarrow Fe(OH)3\downarrow + 3HCOONH_4$	Aerobic and acidi environment,	*Shewanella alga, Shewanella putrefaciens, Stenotrophomonas maltophilia* et al.	Soil reinforcement, anti-seepage plugging, heavy metal pollution control

Urea hydrolysis by ureolytic bacteria

The mechanism of CaCO3 precipitation by **urea hydrolysis**:

$$CO(NH_2)_2 + H_2O \rightarrow 2NH_3 + CO_2$$

$$2NH_3 + 2H_2O \rightarrow 2NH_4{}^{+} + 2OH^{-}$$

$$CO_2 + H_2O \rightarrow HCO_3{}^{-} + H^{+}$$

$$HCO_3{}^{-} + H^{+} + 2OH^{-} \leftrightarrow CO_3{}^{2-} + 2H_2O$$

$$Ca^{2+} + CO_3{}^{2-} \rightarrow CaCO_3(s)\downarrow$$

Application: Consolidation of Sandy soil

- Links the soil particles together - **effective bridging** predominantly concentrated at the point of contacts (**pore throats**).

- The introduction of bacteria to soil can be achieved generally through two main methods including:
  (1) **injection method**
  (2) **premixing method**.

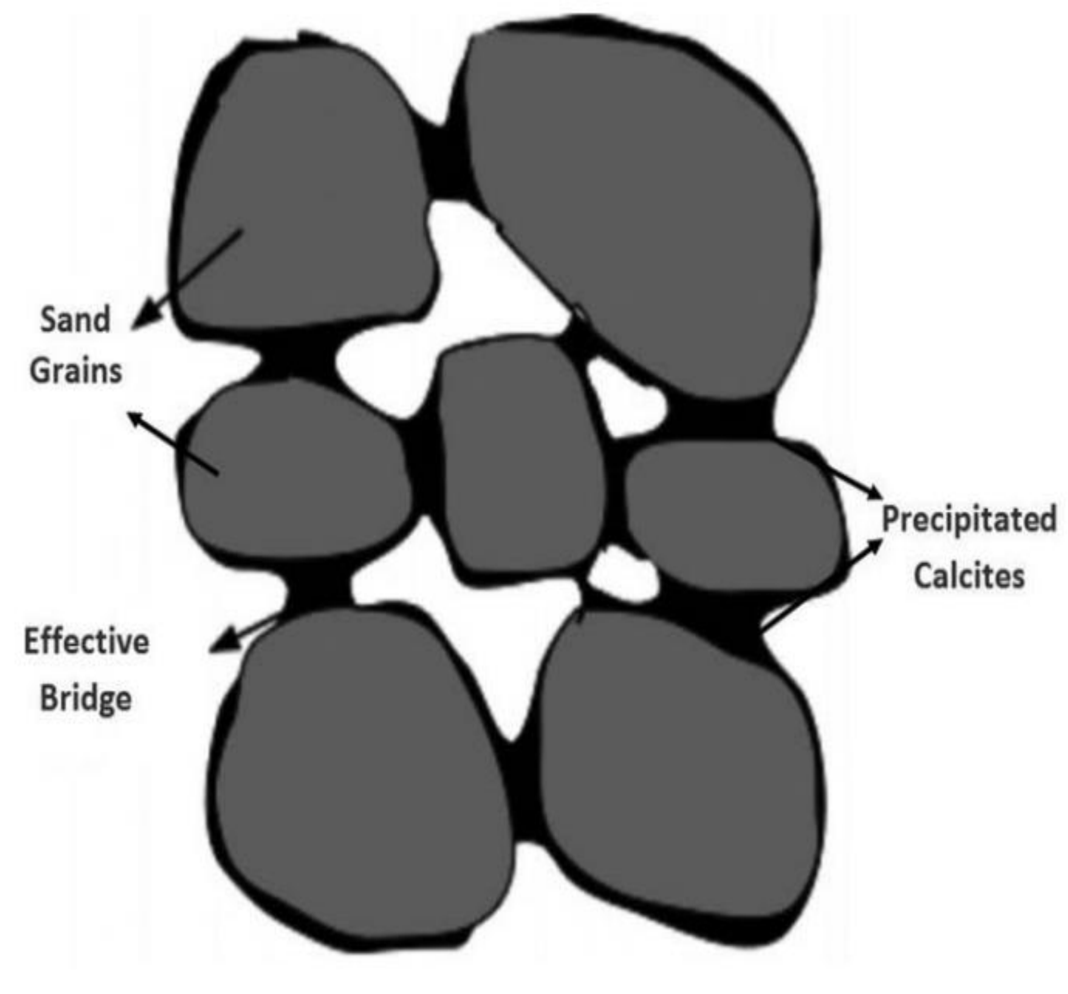

Fig.2. Schematic diagram showing the effective bridge formation (Cheng et al. 2013)

1.Injection Method

- Involves alternating the injection of bacterial culture and cementation solution from top to bottom.

- After injecting bacteria, a **calcium chloride ($CaCl_2$) solution** is added to promote bacterial attachment to sand grains.

- Still the **most commonly preferred** MICP treatment method because
  - Allows control over flow, pressure, hyadraulic gradient.

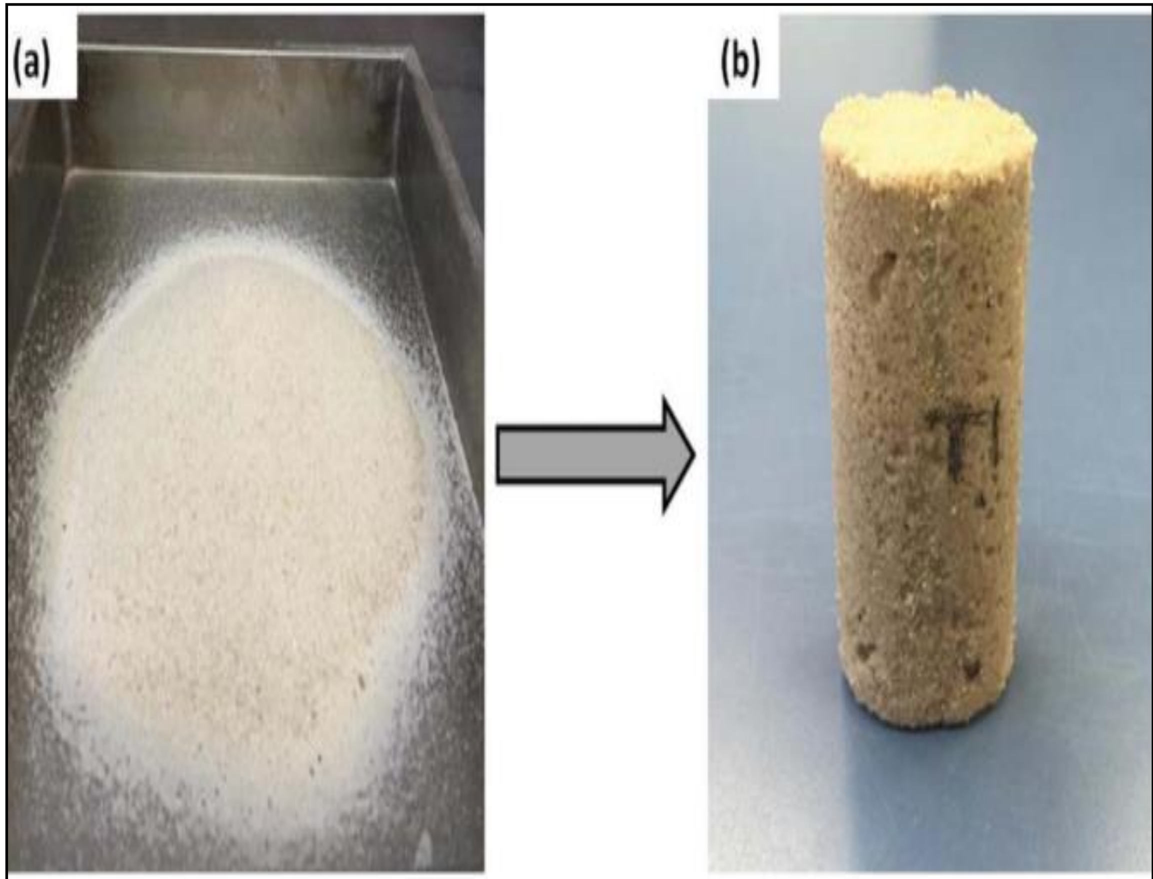

Fig.3. Sand Metamorphosis (a) Natural sand (b) Biocemented sand (biosandstone) (Mujah et al. 2016)

2.Premixing Method

- Involves **mechanically mixing bacteria with soil** until a desired homogeneity is achieved - produces **uniform** $CaCO_3$ distribution throughout.

- The UCS values range from **400 kPa to 1.6 MPa**, lower than other methods but with better uniformity.

- Disturbs the soil- **pseudo stresses -** make mechanical testing results less reliable.

- A modified approach- submerged premixed bacteria-sand in a reactor with cementation solution- **free diffusion** into the sample.

Engineering properties of MICP treated soil

- Increased strength
- Reduced permeability
- Stability against erosion
- Greater ductility and flexibility
- Enhanced soil workability

Application: Self-healing concrete

- Aids in sealing the cracks (self-healing property).

- **Two popular techniques** of incorporating microorganisms into concrete:

  - Direct method - includes **surface spraying**, injecting bacteria, **directly mixing** bacteria with concrete mix.
  - Indirect method – **encapsulation** within silica gel, bio-hydrogel, etc.

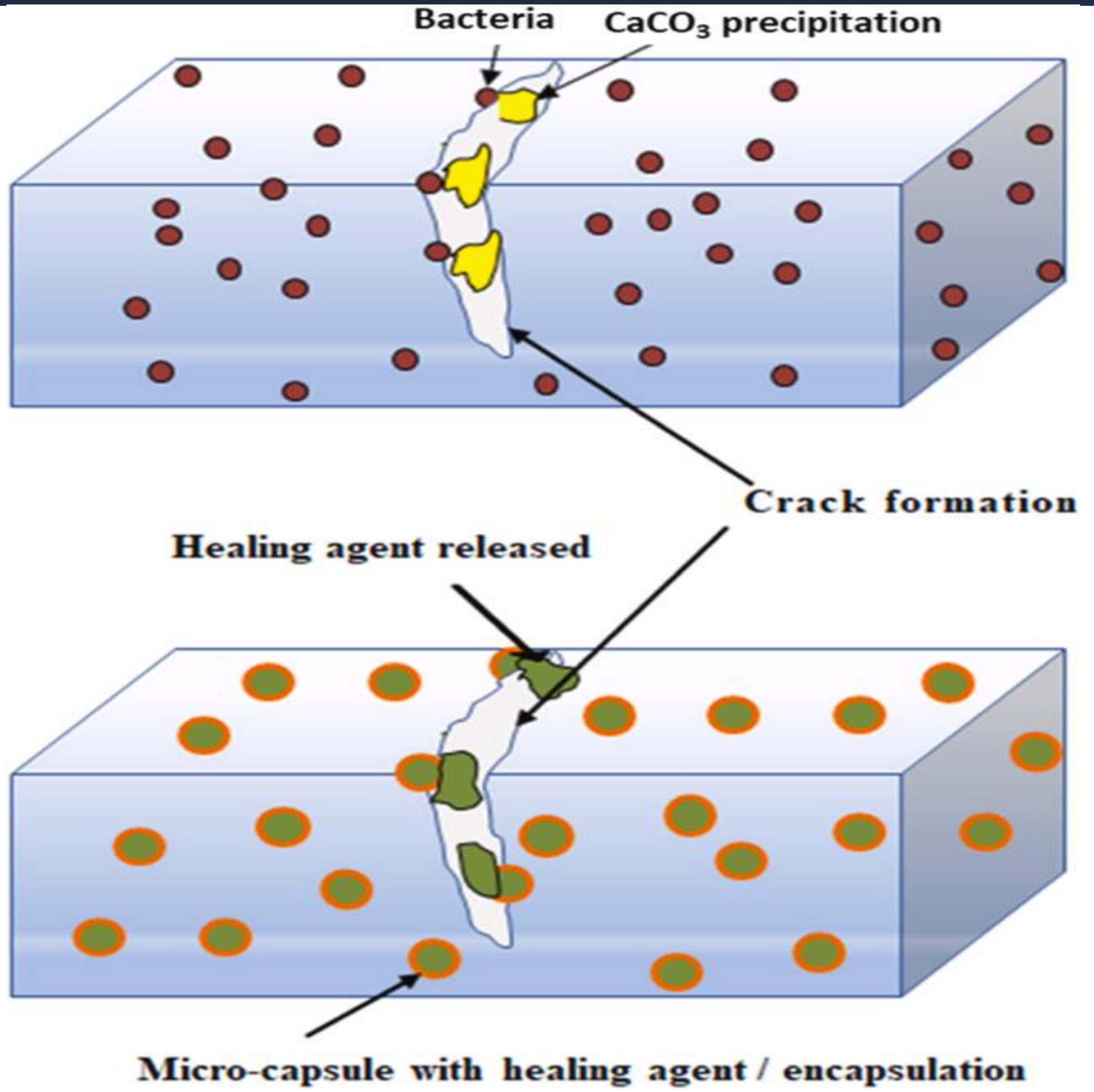

Fig.4. Implementation methods : a) Direct mixing; b) Encapsulation of bacteria (Javeed et al. 2024)

Mechanism

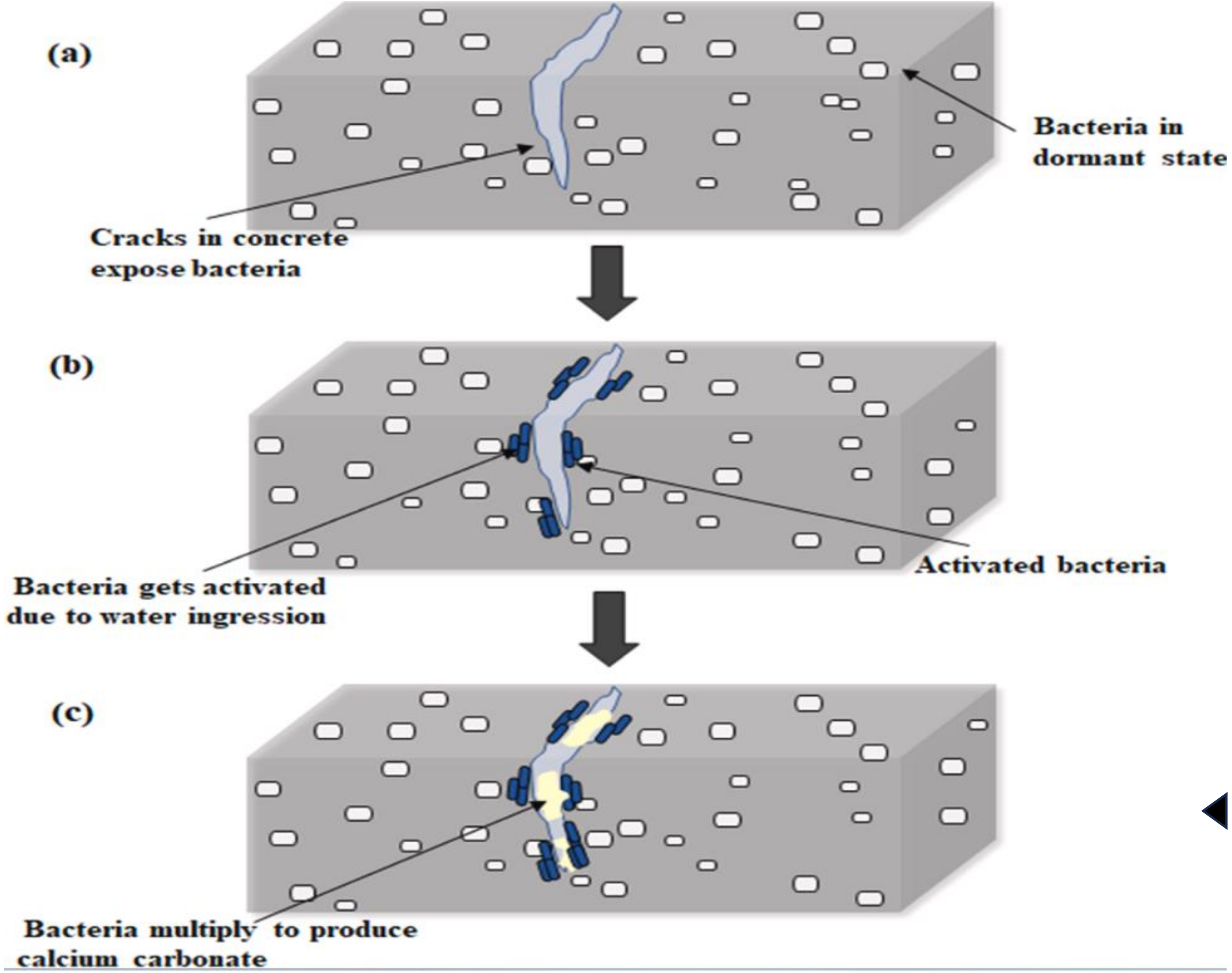

Fig.5. Self-healing mechanism in bacteria concrete
(Javeed et al. 2024)

Application: Bio-remediation of heavy metals

- Remediation of soil and water contaminated with heavy metals (Cd, Cr, Cu, Pb, Ni, Zn), metalloids (As, Hg), ions (NO_3^- and F^-) and radioactive elements (Sr and U).

- **Metal co-precipitation** along $CaCO_3$.

- **Heavy metal sorption** on calcite surfaces.

- pH increase – **precipitation of metal hydroxides**

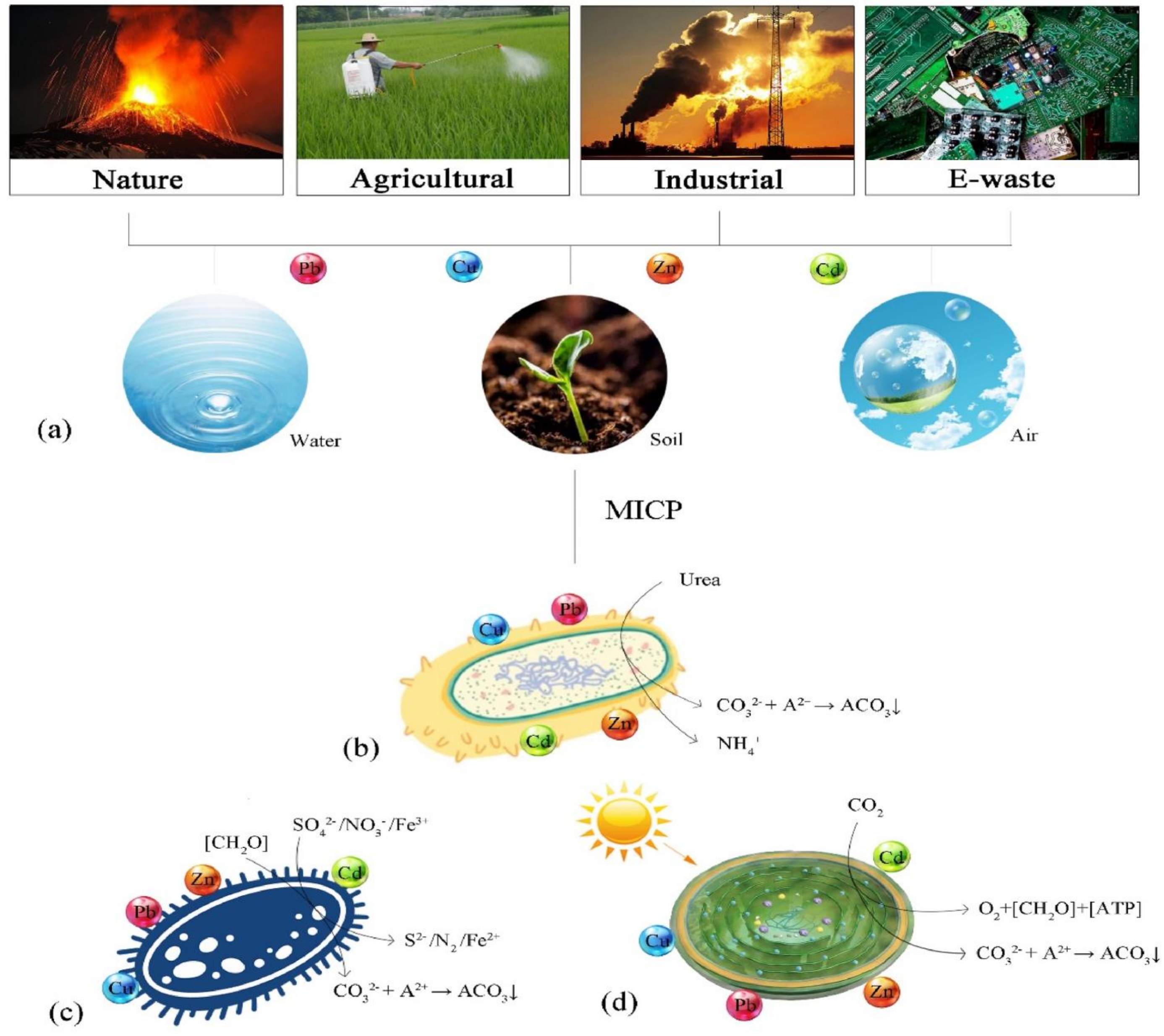

Fig. Sources and mechanisms of carbonate precipitation induced by the microorganism. (Kumar et al. 2023)

Some Other Applications

- Cultural heritage restoration
- Dust remediation
- Soil liquefaction resistance
- Leakage mitigation

Fig. Relationships between Sustainable Development Goals (published by United Nations Department of Global Communications) and the engineering technology (e.g., MICP) (Zhang et al. 2023)

Challenges and Limitations

- Cost of Implementation
- Long treatment time
- High conc. ammonium ions and ammonia as by-products.
- Limited control over uniformity

Future Potential Areas

- Optimization of microbial strains
- Scale-up challenges
- Deep crack repairs
- Long term performance of MICP treated materials
- Tailoring MICP for different soil types

Summary

- MICP- bacteria induce the precipitation of calcium carbonate by altering the chemical environment.

- It improves mechanical properties, reduces permeability, and immobilizes contaminants like metals.

- MICP has broad applications in geotechnical engineering (soil stabilization), self-healing concrete (repairing cracks), bioremediation of heavy metals, carbon sequestration and many more.

References

- Javeed, Y., Goh, Y., Mo, K. H., Yap, S. P., & Leo, B. F. (2024). Microbial self-healing in concrete: A comprehensive exploration of bacterial viability, implementation techniques, and mechanical properties. *Journal of Materials Research and Technology*, *29*, 2376-2395.
- Jiang, L., Xia, H., Wang, W., Zhang, Y., & Li, Z. (2023). Applications of microbially induced calcium carbonate precipitation in civil engineering practice: a state-of-the-art review. *Construction and Building Materials*, *404*, 133227.
- Kumar, A., Song, H. W., Mishra, S., Zhang, W., Zhang, Y. L., Zhang, Q. R., & Yu, Z. G. (2023). Application of microbial-induced carbonate precipitation (MICP) techniques to remove heavy metal in the natural environment: A critical review. *Chemosphere*, *318*, 137894.
- Mujah, D., Shahin, M. A., & Cheng, L. (2017). State-of-the-art review of biocementation by microbially induced calcite precipitation (MICP) for soil stabilization. *Geomicrobiology Journal*, *34*(6), 524-537.
- Zhang, K., Tang, C. S., Jiang, N. J., Pan, X. H., Liu, B., Wang, Y. J., & Shi, B. (2023). Microbial-induced carbonate precipitation (MICP) technology: a review on the fundamentals and engineering applications. *Environmental Earth Sciences*, *82*(9), 229.

THANK YOU